

Informative Power-System Probing via Sequential Bayesian Experimental Design

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Abstract—Active probing provides measurements for identifying uncertain inertia and frequency-response coefficients, but the information obtained depends on the probing signal. Pulse duration affects both the measured transient and the system dynamics preceding later probes. We formulate pulse-duration selection as a sequential Bayesian experimental design problem using noisy endpoint rate-of-change-of-frequency measurements. Six methods are evaluated on reduced IEEE 9-bus and IEEE 14-bus swing models with common sensing, duration constraints, and probe counts. Learned adaptive policies achieve higher mean sequential prior-contrastive (sPCE) information scores than random, optimized fixed-sequence, and myopic designs in both benchmarks. Deep adaptive design (DAD) exceeds the optimized Fixed design by 0.09–0.13 nats on IEEE 9-bus and selects durations through policy forward passes. Step-DAD’s online refinement provides modest additional gains on IEEE 9-bus at substantially greater decision cost, but no consistent mean-score improvement on IEEE 14-bus. Rescoring with more contrasts preserves the learned-policy advantage on IEEE 14-bus, although residual estimator sensitivity limits finer method rankings. These comparisons quantify the information gained and computation required for sequential probing under the specified model and sensing assumptions. Code and saved results are available at https://github.com/alexschmidt123/tpec2027_power_grid.

Index Terms—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

I. INTRODUCTION

The increasing share of inverter-based resources changes the inertia and frequency response of power systems [1]. Characterizing these dynamics requires measurements that distinguish the effects of uncertain parameters. Disturbance-based methods use frequency and active-power responses to estimate inertia [2], but suitable events may not be available when measurements are needed. Active probing supplies deliberate excitation. Its value depends on the signal: field tests on the western North American interconnection show that probing-signal design affects electromechanical-mode estimates [3], and recent work uses probing to estimate inertia and frequency response in systems with inverter-based resources [4].

Excitation design is a central question in system identification [5]. For a pulse applied at a fixed bus and amplitude, duration determines the injected energy and shapes the measured

transient. It also influences the dynamics preceding subsequent probes. Earlier observations can inform later duration choices, so the design problem concerns a sequence of decisions rather than a single pulse. The aim is to reduce parameter uncertainty over the full sequence using the measurements available at each stage.

Sequential Bayesian optimal experimental design (BOED) provides an information objective for this problem [6]. A fixed design optimizes the complete sequence in advance. A myopic design uses observation feedback but considers information from only the next probe. Deep adaptive design (DAD) [7] and reinforcement-learning-based sequential design (RL-sBOED) [8] train policies for a full-sequence objective, shifting optimization to an offline stage. Step-DAD [9] further refines a pretrained policy during the experiment. These approaches differ in both their use of measurements and the computation required before and during probing.

This paper formulates sequential pulse-duration selection for reduced IEEE 9-bus and IEEE 14-bus models with uncertain machine inertia and effective frequency response. Six existing methods are adapted to a common simulator and compared under the same sensing, duration constraints, and probe counts. The physical state evolves across successive probe windows. We examine information scores, selected durations, and offline preparation and online decision costs. Archived IEEE 14-bus histories are also rescored at larger contrast counts to assess the sensitivity of method comparisons to the information estimator. Together, these evaluations establish the information–computation tradeoffs within the specified benchmark.

II. PROBLEM FORMULATION

An episode consists of T probes intended to reduce uncertainty in the grid parameters. We use $t = 1, \dots, T$ for the design stage, d for physical time within a probe window, and D_t for the pulse duration selected at that stage.

A. Latent Parameters and Power-System Model

For a reduced network with m machines, the latent parameters are $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$. Here M_i

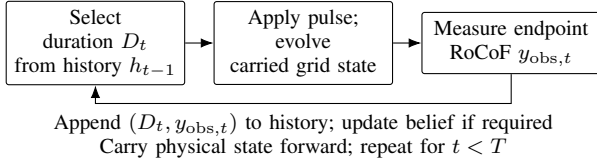


Fig. 1. Sequential probing loop. Adaptive methods use observation history; Fixed and Random select durations without observation feedback.

is an inertia coefficient and K_i is an effective frequency-response coefficient. We assign independent uniform priors, $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$, denoted jointly by $p_0(\theta)$. The parameters are sampled once at the start of an episode and held fixed for all probes.

Let δ_i be rotor angle, ω_i angular-speed deviation, and u_i the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to d . In (1), M_i scales acceleration and K_i models the power response to frequency deviation; no explicit turbine–governor dynamics are included. The damping D_i and couplings B_{ij} are known, and $\Delta f_i = \omega_i/(2\pi)$ is measured in Hz. On a 100-MVA base at 60 Hz, the units of M_i , K_i , and D_i are pu s²/rad, pu/Hz, and pu s/rad. Angles, speed deviations, and net mechanical injections initially equal zero, giving an unforced equilibrium.

The reduced IEEE 9-bus model retains buses 1, 2, and 3; the reduced IEEE 14-bus model retains buses 1, 2, 3, 6, and 8. Test-case reactances [10] define a lossless Laplacian $\mathbf{L}$. Eliminating algebraic buses l while retaining machines g gives $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$ and $B_{ii} = 0$. Resistance, voltage dynamics, and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$, with $f_0 = 60$ Hz. For IEEE 9-bus, $H = (23.64, 6.40, 3.01)$ s matches published machine data [11, Table 2]. For IEEE 14-bus, $H = (4, 6.5, 5, 5, 5)$ s is prescribed at buses (1, 2, 3, 6, 8) as assumed benchmark values. For both networks, $\bar{K}_i = K_{\text{tot}}w_i$, with $K_{\text{tot}} = 0.5$ pu/Hz. The allocation weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE 9-bus and $w = (8, 13, 10, 10, 10)/51$ for IEEE 14-bus. Fixed damping satisfies $D_i = (0.1 \text{ s}^{-1})\bar{M}_i$. The response scales, damping values, and prior ranges are benchmark assumptions rather than validated operational grid parameters.

B. Probe Design and Observation Model

A probe is specified by $\xi_t = (A_t, B_t, D_t)$, comprising its amplitude, application bus, and duration. We fix $A_t = 0.05$

TABLE I
NOMINAL PARAMETER SCALES AND FIXED DAMPING

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	D_i
IEEE 9-bus	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE 14-bus	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

pu and $B_t = 1$ so that duration is the only optimized variable. The applied input is a raised-cosine pulse:

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The pulse acts over $0 \leq d \leq D_t$ and is zero thereafter and at all other buses. Its injected energy is proportional to $A_t D_t/2$, so equal probe counts do not impose equal energy budgets. Each probe occupies a fixed observation window of $D_{\text{obs}} = 3.5$ s. The terminal angles and speeds become the initial state for the next window. Consequently, the response to probe t depends on the preceding durations as well as the current duration, represented by $\xi_{1:t} = (\xi_1, \dots, \xi_t)$.

Let $\Delta f_{1,t}(d; \theta, \xi_{1:t})$ denote the bus-1 frequency deviation during window t . The noise-free scalar response y_t is the signed endpoint rate of change of frequency (RoCoF):

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The parameter and design arguments are suppressed in the numerator. With sampling interval $\Delta d_{\text{samp}} = 0.025$ s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where $\sigma = 0.005$ Hz/s and measurement noise is independent across windows. This sensor measures the post-pulse transient rather than the initial inertial response. Its fixed location and scalar output provide a common observation model for all methods. Observations become available at the end of each window; decision latency is idealized as zero in the simulated dynamics.

C. Sequential Decisions and Bayesian Updating

The data after step t are the ordered design–observation history $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy π selects ξ_t using only observed history h_{t-1} , with amplitude and bus fixed as above. The policy has no access to the generating parameters or the true physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\text{min}}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

Here $\Delta D_{\text{min}} = 0.01$ s. The lower bound is 0.2 s at the first stage and $D_{t-1} + \Delta D_{\text{min}}$ thereafter; the upper bound is $3 - (T - t)\Delta D_{\text{min}}$, reserving room for the remaining

probes. All methods map a unit action continuously into this interval, without sorting durations. Let Π be the resulting policy class. Increasing durations are a benchmark restriction, not a requirement of the swing equations; nonmonotonic sequences are outside this comparison. Policy randomness is independent of θ , and actions depend only on observed history.

For a candidate θ , the predicted response and sensor noise define a Gaussian likelihood $p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t)$ with mean $y_t(\theta, \xi_{1:t})$ and variance σ^2 . Incorporating the new measurement gives

$$p(\theta | h_t) = \frac{p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t) p(\theta | h_{t-1})}{p(y_{\text{obs},t} | h_{t-1}, \xi_t)}. \quad (6)$$

The denominator integrates over θ ; before the first measurement, $p(\theta | h_0) = p_0(\theta)$. Each candidate carries its own state through the observed duration sequence. No policy action probability enters the update: conditional on the observed history, actions depend on neither the unknown parameters nor their candidate states.

D. Expected Information Gain and Design Objective

The expected information gain (EIG) of policy π measures the mean Kullback–Leibler divergence between the terminal posterior and the prior:

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T | \pi} [D_{\text{KL}}(p(\theta | h_T) \| p_0(\theta))]. \quad (7)$$

Here $D_{\text{KL}}(p \| q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$; information gain is measured in nats. The expectation averages over parameter draws, measurement noise, and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

All methods target (8), but their approximations and use of feedback differ. Probe counts are held equal, and computational costs are reported separately.

Evaluating EIG directly requires integration of the nonlinear likelihood over the uncertain parameters. The sequential prior-contrastive (sPCE) lower bound [7] supplies a tractable alternative for training and evaluation. A generating draw $\theta^{(0)} \sim p_0$ produces the observed history, while L independent draws $\theta^{(1)}, \dots, \theta^{(L)} \sim p_0$ provide contrasts. Along the realized design sequence, a candidate's likelihood is

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} | \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

We compute likelihoods in the log domain. The average of $\hat{I}_t$ estimates the sPCE lower bound on $\mathcal{I}_T(\pi)$ rather than EIG itself. A sampled score can be negative and cannot exceed $\log(L+1)$. At $L = 128$, this ceiling is 4.8598 nats and can obscure differences between informative designs. We therefore use the term information score for the estimated bound and

examine contrast-count sensitivity separately. Generating parameters are used by the simulator and scorer, but are never supplied to the policy.

III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). Adaptation and planning horizon are distinct: an adaptive method can be myopic, whereas a fixed sequence can be optimized jointly for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

A. Nonadaptive Designs: Random and Fixed

Random samples $z_t \sim \mathcal{U}(0,1)$ independently at each step and sets $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$. Here $D_1^{\min} = 0.2$, $D_t^{\min} = D_{t-1} + \Delta D_{\min}$ for $t > 1$, and $D_t^{\max} = 3 - (T - t)\Delta D_{\min}$. Each duration is therefore uniform within the remaining feasible interval; complete sequences are not sampled uniformly. Random does not use observations and needs no offline training.

Fixed optimizes all durations before any test observations are collected [6, Sec. 2.4]. Its objective is the full-sequence sPCE score. Validation first compares short, central, long, and spread interior initializations; optimization then proceeds from the selected sequence. A separate sequence is fitted for each training seed. At test time, these durations are executed without measurement feedback.

B. Online One-Step Design: Myopic

Myopic conditions each decision on the available measurements and targets information from the next observation [6, Sec. 2.4]. A weighted particle approximation represents the current posterior. After a measurement, each particle advances its own physical state under the applied duration and receives an updated likelihood weight. The duration search uses three rounds of 16 proposals, with 32 simulated observations per proposal, and returns the best estimated one-step design. It respects the common duration constraints and requires no offline training.

C. Offline-Trained Adaptive Methods: DAD and RL-sBOED

DAD and RL-sBOED learn from sequences simulated under the parameter prior. Their inputs encode the stage index, past durations, scaled observations, and availability masks; future history slots are masked. Once trained, either policy selects a duration with a forward pass followed by the feasible-duration map. Explicit posterior assimilation and candidate-probe simulation are unnecessary during action selection.

The DAD implementation [7] has two 64-unit tanh hidden layers, a trainable stage bias, and a sigmoid action map. Pathwise gradients propagate through observations and carried states to maximize the terminal sPCE score. Validation chooses between random initialization and initialization from a separately optimized Fixed sequence. The reported preparation cost includes the Fixed stage in either case (Section V-C).

Our RL-sBOED implementation [8] uses a REDQ-style actor–critic algorithm with ten critics and five replay updates

per trajectory batch. The actor has two 128-unit ReLU hidden layers. Rewards are $R_t = \hat{I}_t - \hat{I}_{t-1}$, with $\hat{I}_0 = 0$ and discount factor $\gamma = 1$, so their sum equals the terminal sPCE score. Training uses entropy regularization in normalized tanh-action coordinates with target entropy -1 . Evaluation uses the deterministic actor output; reported information scores exclude the entropy bonus.

D. Online Refinement: Step-DAD

Step-DAD [9] begins each episode with the selected DAD checkpoint and retains its first action. After each nonterminal observation, a 128-particle posterior provides parameters and carried states for simulations of the remaining probes. A fresh Adam optimizer takes four updates before the next decision, using batch size 32 and learning rate 10^{-3} . A common conditional validation batch compares the updated policies with the unchanged policy. The selected weights are retained within the episode and reset afterward. This gives $4(T-1)$ refinement updates. Both Myopic and Step-DAD assimilate observations before subsequent decisions; terminal information scoring is a separate operation.

IV. EXPERIMENTAL SETTINGS

A. Simulation and Training Protocol

We evaluate horizons $T = 3, 4, 5$, with one scalar observation per probe. Policies are trained separately for each network and horizon; sensor settings and duration constraints are shared.

Simulation uses fourth-order Runge–Kutta (RK4) integration with a $1/640$ -s step. The pathwise gradient calculation approximates state and duration Jacobians numerically: perturbations are $10^{-5} \max(1, |x_j|)$ for state coordinate x_j and 10^{-5} s for duration, with duration perturbations clipped at the feasible bounds. Policy and conditional-refinement gradients are checked against end-to-end finite differences in the validation suite.

Each network–horizon combination has Fixed, DAD, and RL-sBOED fits from training seeds 101, 202, and 303. Training comprises 2000 trajectory updates of batch size 32. Fixed and DAD use Adam at 10^{-3} ; the RL actor and critics use 3×10^{-4} . Every 100 updates, 128 validation systems determine checkpoint selection. Training, validation, and initial evaluation use $L = 128$ contrasts. The larger- L IEEE 14-bus evaluation is described in Section V-D. Myopic and Step-DAD each use 128 planning particles, and Step-DAD takes four refinement updates after each nonterminal observation.

B. Evaluation Protocol

Test seeds 1001, 1002, and 1003 supply 512 episodes each, or 1536 episodes per fitted policy. Methods share paired parameter and measurement-noise draws; planner randomness uses a separate stream. Training, checkpoint validation, and testing also use separate seeds. For fitted methods, tables report the mean and sample standard deviation (SD) of three training-seed means, each averaged over all test episodes. Random and Myopic do not depend on a training seed, so their SDs

are taken over the three test-seed means. These SDs quantify variation among seed means. They neither describe episode dispersion nor provide uncertainty intervals for paired method differences.

V. RESULTS AND DISCUSSION

A. Information Acquired at Equal Probe Budgets

Tables II and III report sPCE information scores using the aggregation defined in Section IV-B. All three learned adaptive methods exceed Random, Fixed, and Myopic in mean score for each network and horizon.

TABLE II
IEEE 9-BUS sPCE SCORES ($L = 128$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

TABLE III
IEEE 14-BUS sPCE SCORES ($L = 128$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

The IEEE 9-bus DAD–Fixed mean differences are 0.1253, 0.0903, and 0.1286 nats for $T = 3, 4, 5$. Step-DAD adds a further 0.0105, 0.0287, and 0.0269 nats to DAD. All scores are below the 4.8598-nat ceiling. The comparisons favor the learned full-sequence policies in this benchmark, although they do not separate the contribution of observation feedback from that of policy parameterization and optimization. The mean differences are descriptive; separate method SDs are not used to claim statistical significance.

IEEE 14-bus presents a different estimation regime: the learned-policy means at $L = 128$ approach the 4.8598-nat ceiling. Step-DAD differs from DAD by -0.0049 , -0.0028 , and $+0.0003$ nats for $T = 3, 4, 5$, which does not establish an EIG ranking. Fixed exceeds Myopic at $T = 4, 5$ in Table III, but the archived-history rescoring reverses that ordering (Section V-D).

B. Selected Probe Durations

Figure 2 summarizes duration distributions pooled across test episodes and, for fitted policies, training seeds. Fixed and the learned methods contribute 4608 histories each; Random and Myopic contribute 1536 each. Bars show interquartile

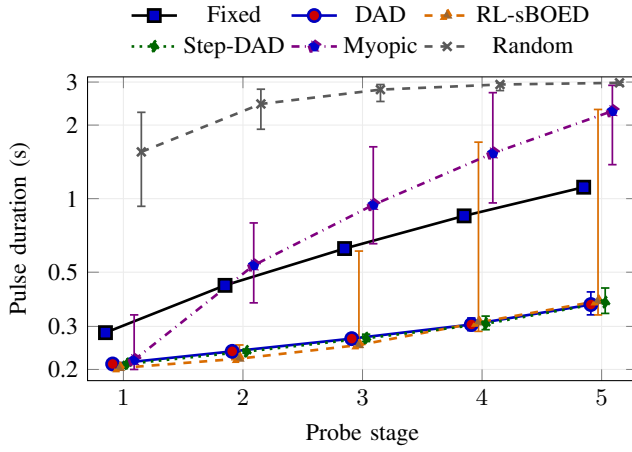


Fig. 2. IEEE 14-bus probe durations at $T = 5$. Markers show medians; bars span the 25th–75th percentiles. The vertical axis is logarithmic; horizontal offsets separate methods.

ranges, not confidence intervals. At $T = 5$, final-stage medians are 0.369 s for DAD, 0.376 s for Step-DAD, 1.114 s for Fixed, 2.283 s for Myopic, and 2.979 s for Random. Sequential uniform sampling pushes Random’s later durations toward the upper bound. RL-sBOED’s final-stage medians are 2.585, 0.381, and 0.324 s for training seeds 101, 202, and 303, respectively; this variation contributes to its wide pooled ranges. All sequences obey the imposed ordering constraint, and different durations also imply different injected energies.

C. Computational Cost

Table IV reports timings on NVIDIA A100 PCIe 40-GB GPUs, using training seed 101 and test seed 1001. Offline preparation includes initialization, optimization, and checkpoint validation; DAD and Step-DAD include both Fixed and DAD training. Online time is the mean total decision time per sequence over 512 test episodes per method. It includes action selection and, where applicable, posterior assimilation and refinement; true-system propagation and information scoring are excluded. IEEE 9-bus profiling excludes one warm-up episode per method. Offline times are single-run measurements. For all six methods (Random, Fixed, Myopic, DAD, RL-sBOED, and Step-DAD), the timed runs use the same contrast count, $L = 128$, for information scoring. Training, validation, and Step-DAD refinement also use $L = 128$; Myopic uses 128 posterior particles and 32 simulated observations per candidate. Increasing L increases evaluation scoring time, which is excluded from the decision timers.

Let C_s denote one candidate-window simulation cost and C_π a policy forward pass. Dense m -machine RK4 integration over N time steps gives $C_s = O(Nm^2)$.

Online policy work is $O(T)$ for Random/Fixed and $O(TC_\pi)$ for DAD/RL-sBOED. Myopic uses $O(TQC_e + TPC_s)$, where $P = 128$ particles, $Q = 48$ candidate evaluations per decision, and C_e includes candidate simulation and likelihood calculations. Step-DAD refinement simulates $UB(L+1)T(T-1)/2$ candidate windows before accounting for numerical Jacobians

and conditional validation, with $U = 4$, $B = 32$, and $L = 128$. Posterior assimilation adds $O(TPC_s)$ work.

TABLE IV
COMPUTATIONAL COSTS FOR BOTH IEEE NETWORKS

Method	IEEE 9-bus			IEEE 14-bus		
	$T = 3$	$T = 4$	$T = 5$	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)						
Random	0	0	0	0	0	0
Fixed	0.554	0.802	1.063	1.880	2.750	3.720
Myopic	0	0	0	0	0	0
DAD	1.116	1.608	2.139	3.810	5.457	7.418
RL-sBOED	0.100	0.117	0.129	0.163	0.205	0.249
Step-DAD	1.116	1.608	2.139	3.810	5.457	7.418
Online decisions (ms per sequence)						
Random	0.167	0.229	0.303	0.158	0.207	0.285
Fixed	0.849	1.135	1.472	1.141	1.906	1.952
Myopic	4.418	5.952	7.514	9.998	13.433	16.918
DAD	1.012	1.345	1.745	1.440	1.513	1.733
RL-sBOED	1.147	1.781	7.692	1.413	1.550	1.905
Step-DAD	3.083	7.689	14.346	9.245	23.320	45.089

The leading simulation and gradient cost of Fixed and DAD is $O(JB(L+1)TC_g)$ for J updates, batch size B , and per-window gradient cost C_g . DAD and Step-DAD preparation includes both stages, $W_F + W_D$, where W_F and W_D are the Fixed and DAD training times. RL-sBOED also incurs actor-critic replay updates. Random and Myopic require no offline training. The online tradeoff is substantial: at IEEE 9-bus $T = 5$, Step-DAD adds 0.0269 nats to DAD’s mean score but takes 14.346 s rather than 1.745 ms per sequence.

Information evaluation has its own cost. Rescoring n archived histories requires $O(n(L+1)TC_s)$ simulation work and $O(nLT)$ likelihood arithmetic, without retraining or selecting new actions. The decision timers characterize action selection; complete evaluation runtimes also include system propagation and information scoring.

D. Estimator Sensitivity

The sampled sPCE score has ceiling $\log(L+1)$, whereas true EIG has no such contrast-count ceiling. We rescore all 64 512 distinct IEEE 14-bus histories at $L = 10^4$ and 10^5 , holding policies, generating parameters, durations, and observations fixed. No policies are retrained or rerun. Contrasts are drawn from the archived prior; matching test-seed/episode pairs share the contrast stream across methods, and the 10^4 samples are a prefix of the 10^5 samples. Archived terminal states and original scores are reproduced before rescoring. Tables V and VI report mean $\pm$ SD using the seed aggregation defined in Section IV-B; seed-level summaries are included in the repository.

At $L = 10^4$ and 10^5 , the score ceilings are 9.2104 and 11.5129 nats. Each learned policy retains a higher mean score than the three baselines. Myopic exceeds Fixed at every horizon, reversing their $L = 128$ ordering for $T = 4, 5$. That

TABLE V
IEEE 14-BUS SPCE SCORES ($L = 10^4$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.8449 ± 0.0121	0.9552 ± 0.0162	0.9991 ± 0.0388
Fixed	5.2956 ± 0.0802	6.0609 ± 0.1246	6.7906 ± 0.0167
Myopic	5.8136 ± 0.0514	6.4715 ± 0.0127	6.8192 ± 0.0500
DAD	7.1062 ± 0.2327	8.2061 ± 0.0286	8.6662 ± 0.0556
RL-sBOED	7.0793 ± 0.0567	8.1296 ± 0.1355	8.5852 ± 0.2660
Step-DAD	7.1008 ± 0.2309	8.1992 ± 0.0191	8.6583 ± 0.0550

TABLE VI
IEEE 14-BUS SPCE SCORES ($L = 10^5$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.8476 ± 0.0136	0.9645 ± 0.0150	1.0075 ± 0.0333
Fixed	5.3472 ± 0.0837	6.1694 ± 0.1401	7.0488 ± 0.0272
Myopic	5.9348 ± 0.0697	6.7628 ± 0.0277	7.2307 ± 0.0540
DAD	7.3126 ± 0.2674	8.8217 ± 0.0707	9.6931 ± 0.1308
RL-sBOED	7.2873 ± 0.0578	8.7141 ± 0.2070	9.5528 ± 0.4753
Step-DAD	7.3064 ± 0.2650	8.8135 ± 0.0578	9.6947 ± 0.1220

reversal demonstrates finite-contrast sensitivity; it does not identify ceiling saturation as the sole cause. At $L = 10^5$, Step-DAD differs from DAD by -0.0062 , -0.0082 , and $+0.0016$ nats for $T = 3, 4, 5$, so refinement has no consistent mean benefit on IEEE 14-bus. DAD's $T = 5$ mean rises from 8.666 to 9.693 nats between the two larger contrast counts. The additional contrasts alleviate the original ceiling effect, but the residual change prevents a claim of convergence or a definitive EIG ranking among learned policies.

E. Scope and Limitations

These results concern reduced lossless dynamics, specified parameter priors, increasing pulse durations, and a single fixed actuator and endpoint sensor. The sPCE score estimates a lower bound on expected joint uncertainty reduction; individual parameter-estimation accuracy is not evaluated. Training and testing use the same prior, leaving prior shift, sensor bias, and unmodeled dynamics outside the present assessment. Finite contrasts, posterior particles, and numerical integration introduce separate approximations. Finally, zero decision latency is assumed in the simulated dynamics, so the measured algorithm costs alone do not establish real-time feasibility or improved control safety.

VI. CONCLUSION

Selecting a probing duration changes the immediate measurement and the state available to later experiments. We formulated this coupled design problem for uncertain inertia

and effective frequency response and compared six methods on reduced IEEE 9-bus and IEEE 14-bus models. Learned adaptive policies obtain higher mean sPCE scores than the tested baselines at equal probe counts. DAD and RL-sBOED place optimization offline, while Step-DAD spends additional online computation for modest IEEE 9-bus gains. On IEEE 14-bus, larger- L evaluation retains the learned-policy advantage, reverses the longer-horizon Fixed–Myopic ordering, and shows no consistent refinement benefit. Residual estimator sensitivity limits finer EIG comparisons. The results support further investigation of sequential probing under model mismatch and practical sensing and latency constraints.

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